

SAT-SA

Supervisory Analytics Tool for SOC Assessment

Smart India Hackathon · NCIIPC problem statement · Synthetic offline prototype

Periodic SOC evidence → explainable review indicators → source records → human decision

Find which Critical Sector Entities and samples deserve examination, with a reason an examiner can check.

Not a SOC, SIEM, real-time monitoring platform, or automated compliance decision maker.

Architecture and evidence boundary

Web workspace	Next.js and TypeScript present assessment context, filters, source drill-down, report and human review.
API and workflow	FastAPI validates imports, owns jobs and identity, serves paginated records and records audit events.
Analytics	Versioned Python rules evaluate eight supervisory families, negative space, matched peers and review priority.
Local storage	Immutable Parquet evidence and DuckDB metadata preserve provenance, run configuration and decisions.

Docker Compose runs locally without runtime Internet access. Ground-truth labels live outside analytical inputs and normal evidence APIs.

Negative space is an explicit review question

Declared monitoring expectations establish which critical assets should be represented in submitted evidence.

The result states expected, observed and missing records, plus confidence and limitations.

No alert activity is a proxy. It does not prove a monitoring control failed.

Assessment run

Dataset

demo-2024 - 2026-09-29T16:08:50.490652+00:00

demo

Run analytics

Refresh workspace

Period: 2024-01-01T00:00:00Z — 2025-01-01T00:00:00Z · Dataset hash: 574def4e7a2ef971 · Analytics 1.1.0

Overview

Entities

Review queue

Signals

Negative space

Period comparison

Report

Data ingestion

Validation

Audit trail

Negative space

CSE filter

All entities

Name	Category	Severity	Evidence Count	Confidence	Observed Evidence	Examine
Expected escalation evidence absent	escalation	high	144	limited	144 of 440 eligible alerts match the documented predicate.	Inspect evidence
Expected critical-asset activity absent	cyber_resilience	high	4	limited	4 of 8 eligible assets match the documented predicate.	Inspect evidence
Low activity relative to comparable entities	cyber_resilience	medium	1	limited	1 of 1 eligible cses match the documented predicate.	Inspect evidence
Expected escalation evidence absent	escalation	high	576	limited	576 of 744 eligible alerts match the documented predicate.	Inspect evidence

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Analytics will identify indicators for human examination. They will not make automatic supervisory findings.

SAT-SA prototype · Synthetic evidence · Indicators require human examination

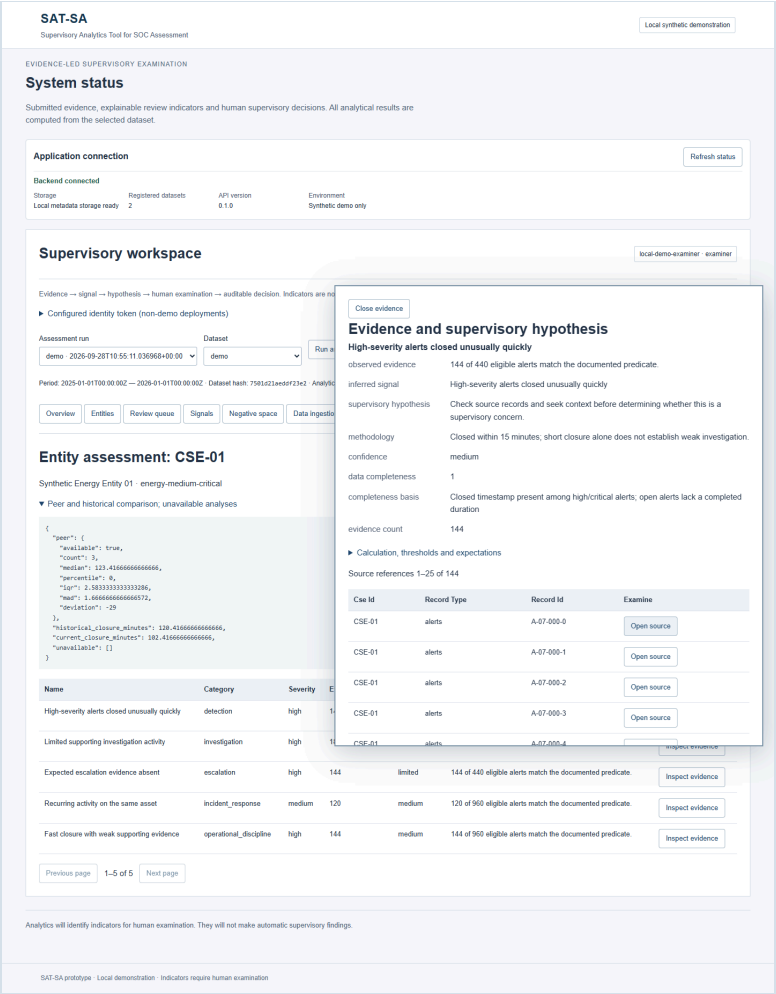
Actual local demo: four assets have no submitted alert activity; examiner can inspect the source references.

From indicator to source evidence

The examiner sees the observation, predicate, calculation, completeness and supervisory hypothesis.

References resolve to entity-scoped normalized records with source provenance.

A human records an outcome and note. The audit trail preserves that action without rewriting the analytics.



Actual demo capture. Indicators are prompts for review, not findings.

Submission evidence and next validation gate

The deterministic demo covers 8 CSEs, 2 sectors, 6,816 alerts, 852 cases and 192 assets.

Tests cover immutable versions, relationship integrity, analytical rules, API workflow, browser flow and disconnected runtime.

Expert manual-review comparison remains unmeasured. Million-record performance and production accreditation are open gates.

Print or save as PDFDownload source report JSON

Analytical indicators are review prompts, not findings of non-compliance. Missing evidence does not prove control failure.

Run 20260929T160850-3e1c7b185ded · Dataset demo-2024 · Hash 574def4e7a2af971d5316aab7fc03de17b77e9dbd127cc977167a95e4ae0a866

Assessment scope

First 50 signals, first 10 references per signal, and first 20 suggested samples. Use the evidence API for complete records.

8 entities · 25 indicators · 2329 suggested review samples · 0 recorded decisions

Entities

CSE	Sector	Alerts	Cases	High-Priority Indicators	Closure Completeness
CSE-01	energy	960	120	4	1
CSE-02	energy	960	120	1	1
CSE-03	energy	960	120	1	1
CSE-04	energy	960	120	0	1
CSE-05	financial_services	960	120	0	1
CSE-06	financial_services	96	12	0	1
CSE-07	financial_services	960	120	0	1
CSE-08	financial_services	960	120	4	1

Review indicators and source references

CSE-01 · Fast closure with weak supporting evidence

144 of 960 eligible alerts match the documented predicate. Confidence: medium. Evidence records: 144.

Method: Intersection of fast closure and limited submitted investigation events; no inference of intent.

Sample references: CSE-01/alerts/A-07-000-0, CSE-01/alerts/A-07-000-1, CSE-01/alerts/A-07-000-2, CSE-01/alerts/A-07-000-3, CSE-01/alerts/A-07-000-4, CSE-01/alerts/A-07-000-5, CSE-01/alerts/A-07-000-6, CSE-01/alerts/A-07-000-7, CSE-01/alerts/A-07-001-0, CSE-01/alerts/A-07-001-1

Selected-run report can be printed or downloaded as JSON with run hash and evidence references.